VibeGuard Cloud: Load-Shift-Aware Industrial Motor Bearing Fault Classification Using 1D Deep Learning and AWS

Final-Year Academic Project Report | Official Run ID: **vibeguard-20261102T092858Z-s42** | Repository: **shaziyasultanablr-byte/Cloud-Computing-Final**

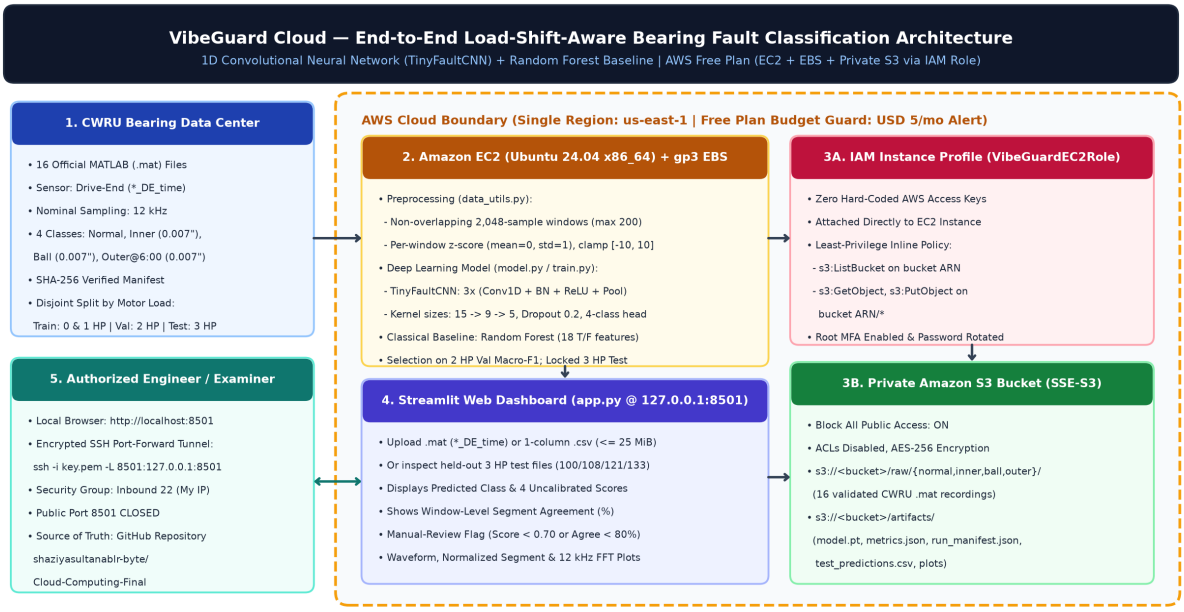
1. Abstract

Unexpected rolling-element bearing failures in industrial induction motors cause costly unplanned downtime. While deep learning models applied to accelerometer signals often report high accuracy on laboratory datasets, many studies rely on random window-level splits that leak identical operating loads across train and test sets. This project presents **VibeGuard Cloud**, an end-to-end 1D deep learning and AWS cloud engineering prototype that evaluates **cross-load bearing fault generalization** using 16 official drive-end (**_DE_time*) recordings from the Case Western Reserve University (CWRU) Bearing Data Center. Models are trained exclusively on 0 HP and 1 HP recordings (8 files, 673 non-overlapping 2,048-sample windows), selected on 2 HP validation recordings (4 files, 377 windows), and evaluated once on locked 3 HP test recordings (4 files, 378 windows) across four classes: Normal, 0.007-in Inner Race, 0.007-in Ball, and 0.007-in Outer Race @ 6:00. During dataset validation, our pipeline discovered and remediated an upstream workspace contamination in CWRU's *99.mat* file (which embeds both *X098_DE_time* and *X099_DE_time*). On the locked 3 HP test split, both the 1D TinyFaultCNN (15,572 parameters) and an 18-feature Random Forest baseline achieve 100.00% window-level and recording-level accuracy; however, Random Forest suffers noticeable confidence dilution on 3 HP Inner Race (*108.mat*: mean score 0.7273, margin 0.5005) whereas 1D TinyFaultCNN maintains a 0.9953 mean uncalibrated score (margin 0.9932).

2. Problem Statement, Objectives, and Literature Context

- **Problem Statement:** Window-level random splitting leaks recording sessions across splits; motor load shifts alter shaft speed and bearing defect frequencies; and ad-hoc cloud deployments frequently expose credentials or public ports.
- **Objective 1 (Data Provenance & Leakage Control):** Validate all 16 CWRU *.mat* recordings via SHA-256 checksums and enforce disjoint motor-load splits (0 & 1 HP train, 2 HP validation, 3 HP test).
- **Objective 2 (1D Deep Learning vs. Classical Baseline):** Train a lightweight PyTorch 1D CNN (*TinyFaultCNN*) on per-window z-scored 2,048-sample segments and compare against a 200-tree Random Forest using 18 time/frequency features.
- **Objective 3 (Decision-Support Manual Review Flag):** Flag recordings when top uncalibrated softmax score < 0.70 or window agreement < 80%, clearly distinguishing softmax scores from calibrated probabilities.
- **Objective 4 (Secure AWS Free Plan Deployment):** Deploy on Ubuntu 24.04 x86_64 EC2 + gp3 EBS + private S3 (Block Public Access ON) via least-privilege IAM role (*VibeGuardEC2Role*) and encrypted SSH port forwarding.
- **Literature Foundation:** Builds upon Smith & Randall (2015) CWRU benchmark analysis, Zhang et al. (2017) wide-kernel 1D CNNs (WDCNN), and Ince et al. (2016) compact 1D motor fault CNNs.

3. Proposed System & AWS Cloud Architecture



The AWS architecture enforces strict Free Plan cost guardrails (USD 5.00 monthly budget alert, single region *us-east-1*, zero paid services) and least-privilege security. The EC2 instance accesses private S3 prefixes (*raw/* and *artifacts/*) exclusively via the attached *VibeGuardEC2Role* instance profile without static access keys. The Streamlit application binds to *127.0.0.1:8501* and is demonstrated over an encrypted SSH tunnel (*ssh -L 8501:127.0.0.1:8501*).

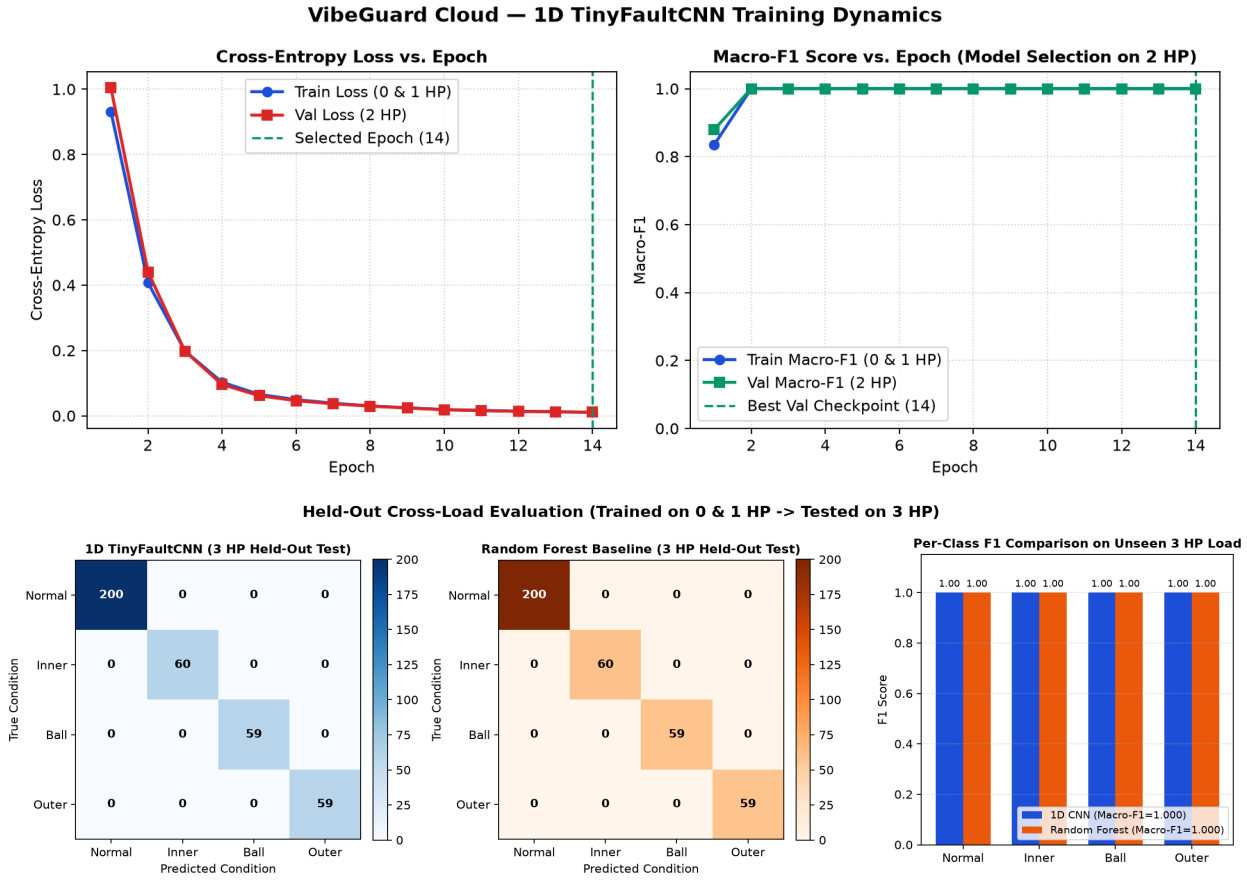
4. Dataset Provenance & CWRU 99.mat Remediation

All 16 official CWRU recordings were verified via *scripts/validate_dataset.py*. Notably, inspection revealed that CWRU's official *99.mat* (2 HP Normal validation) contains both *X098_DE_time* (483,903 samples from 1 HP training *98.mat*) and *X099_DE_time* (485,063 samples). Our loader explicitly matches *X{file_id:03d}_DE_time* to prevent train-to-validation leakage.

ID	File	Class	Load	Split	Channel Key	Samples	Win	SHA-256 (First 16 Hex)
97	normal/97.mat	normal	0 HP	train	X097_DE_time	243,938	119	16bf48babcf1c7ac
98	normal/98.mat	normal	1 HP	train	X098_DE_time	483,903	200	37e6612c05e65c41
99	normal/99.mat	normal	2 HP	val	X099_DE_time	485,063	200	4b97e6b5361f45ef
100	normal/100.mat	normal	3 HP	test	X100_DE_time	485,643	200	88a5990cb541320e
105	inner/105.mat	inner	0 HP	train	X105_DE_time	121,265	59	f80b0ea04fd06b37
106	inner/106.mat	inner	1 HP	train	X106_DE_time	121,991	59	e5cec7cdd138e6cd
107	inner/107.mat	inner	2 HP	val	X107_DE_time	122,136	59	111ba8996a115684
108	inner/108.mat	inner	3 HP	test	X108_DE_time	122,917	60	d415f0e65128bfa0
118	ball/118.mat	ball	0 HP	train	X118_DE_time	122,571	59	b00628f8dd8d1d93
119	ball/119.mat	ball	1 HP	train	X119_DE_time	121,410	59	bc72d9df7668219e
120	ball/120.mat	ball	2 HP	val	X120_DE_time	121,556	59	e0b7a584c49af523
121	ball/121.mat	ball	3 HP	test	X121_DE_time	121,556	59	52f686e984ba8e9b
130	outer/130.mat	outer	0 HP	train	X130_DE_time	121,991	59	35a095307d097147
131	outer/131.mat	outer	1 HP	train	X131_DE_time	122,426	59	7883f7b83beadc54
132	outer/132.mat	outer	2 HP	val	X132_DE_time	121,410	59	17a69ed5d2270b42
133	outer/133.mat	outer	3 HP	test	X133_DE_time	122,571	59	53f076cb0d905cf4

5. Deep Learning Methodology & Experimental Evaluation

Preprocessing: Non-overlapping 2,048-sample windows (stride 2,048, max 200 windows/file), per-window z-score normalization (zero mean, unit variance), clamped to [-10.0, 10.0].
1D TinyFaultCNN (15,572 parameters): Conv1D(1->16, k=15) + BN + ReLU + MaxPool(4); Conv1D(16->32, k=9) + BN + ReLU + MaxPool(4); Conv1D(32->64, k=5) + BN + ReLU + AdaptiveAvgPool(1); Dropout(0.2) + Linear(64->4). Trained for 14 epochs (Adam lr=0.001, batch_size=64, seed=42).
Baseline: Random Forest (200 trees, max_depth=12, seed=42) trained on 18 time/frequency features.

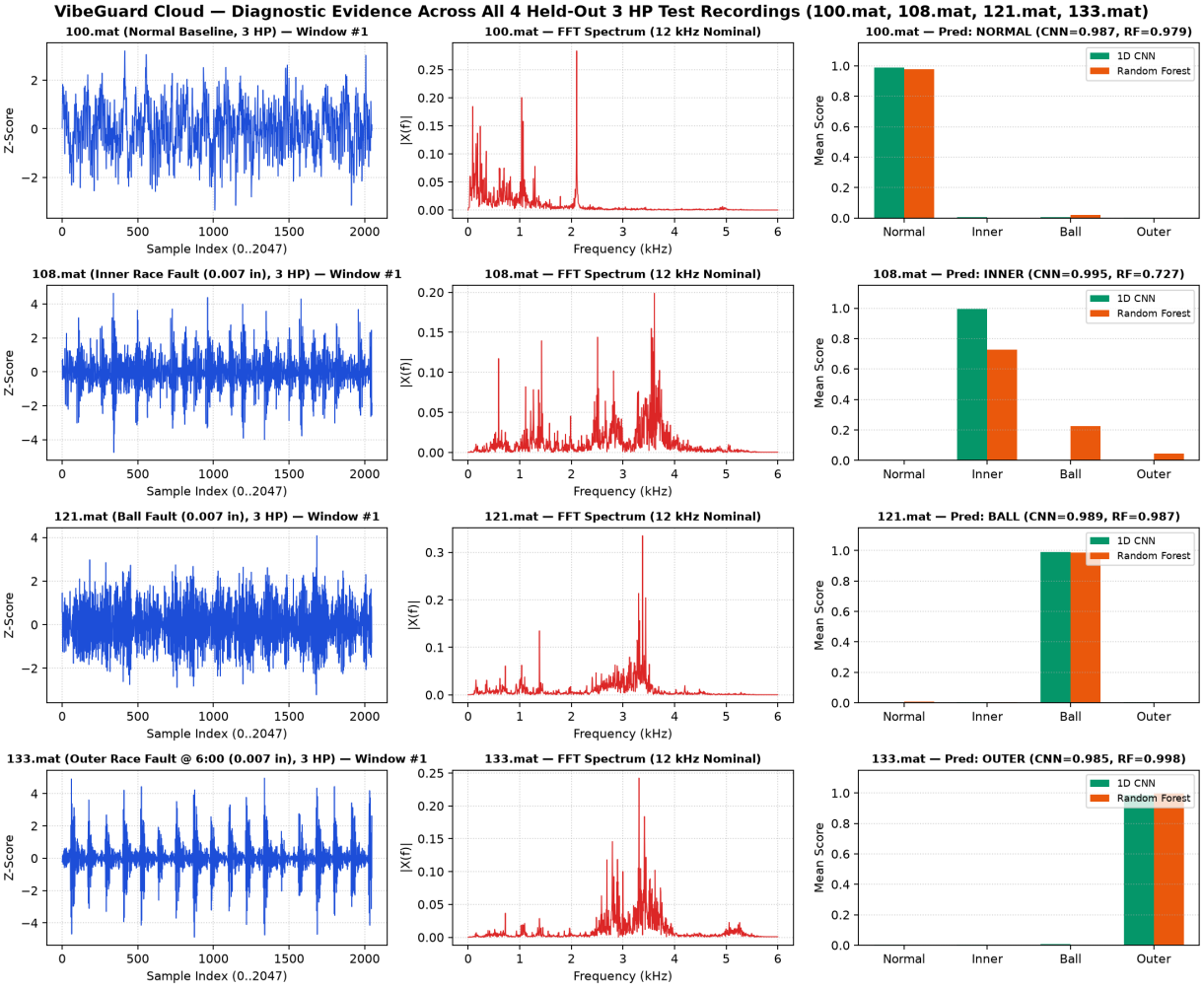


6. Measured Results & Recording-Level Analysis (2 HP Val & 3 HP Test)

Split	File ID	True Class	Windows	CNN Pred	CNN Top Score	CNN Margin	RF Pred	RF Top Score	RF Margin	Review Flag
Val 2HP	99	normal	200	normal	0.9879	0.9834	normal	0.9793	0.9600	False
Val 2HP	107	inner	59	inner	0.9923	0.9886	inner	0.9877	0.9756	False
Val 2HP	120	ball	59	ball	0.9889	0.9824	ball	0.9823	0.9652	False
Val 2HP	132	outer	59	outer	0.9908	0.9863	outer	1.0000	1.0000	False
Test 3HP	100	normal	200	normal	0.9869	0.9820	normal	0.9785	0.9576	False
Test 3HP	108	inner	60	inner	0.9953	0.9932	inner	0.7272	0.5005	False
Test 3HP	121	ball	59	ball	0.9890	0.9840	ball	0.9870	0.9794	False
Test 3HP	133	outer	59	outer	0.9851	0.9777	outer	0.9981	0.9972	False

Discussion of Load-Shift Behavior: Both models classify all 378 windows (100.00%) and all 4 held-out 3 HP recordings accurately because CWRU 0.007-in EDM faults were recorded on the same physical bearing installation across 0–3 HP (~3.7% shaft speed change). However, on *108.mat* (3 HP Inner Race), the Random Forest baseline experiences pronounced confidence erosion: its mean score drops to **0.7273** (with 22.68% of trees voting for Ball fault, margin = 0.5005), barely clearing the 0.70 review threshold. In contrast, 1D TinyFaultCNN retains a **0.9953** mean uncalibrated score (margin = 0.9932).

7. Diagnostic Evidence on Held-Out 3 HP Test Recordings



8. Methodological Limitations, Conclusion, and References

• **Small Recording Count per Load:** Validation (2 HP) and held-out test (3 HP) each contain 1 recording per class (4 files total per split). Window accuracy (378 windows) measures within-recording consistency rather than multi-unit variance.

• **Single Laboratory Rig & Seeded EDM Faults:** CWRU defects are artificial 0.007-in EDM trenches on one test stand; results do not establish generalization across different motor frames or natural fatigue wear.

• **Uncalibrated Softmax Outputs:** Reported scores are uncalibrated softmax activations, not calibrated probabilities.

• **References:** (1) CWRU Bearing Data Center, engineering.case.edu/bearingdatacenter; (2) Smith & Randall (2015), MSSP 64-65, 100-131; (3) Zhang et al. (2017), Sensors 17(2), 425; (4) Ince et al. (2016), IEEE TIE 63(11), 7067-7075.